

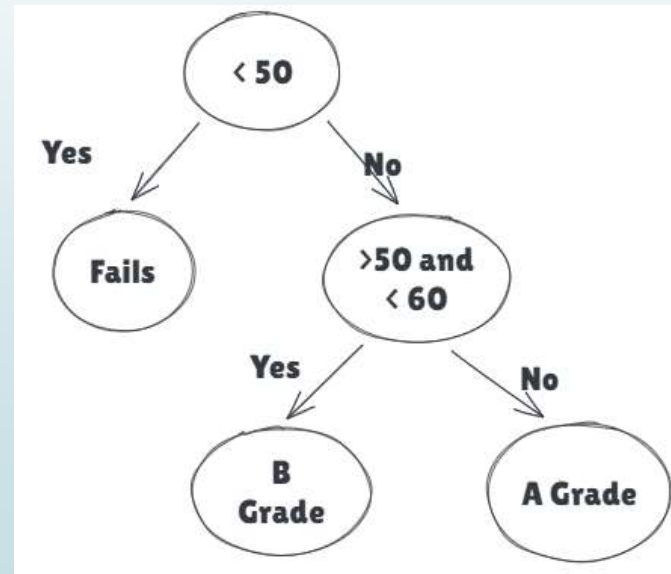


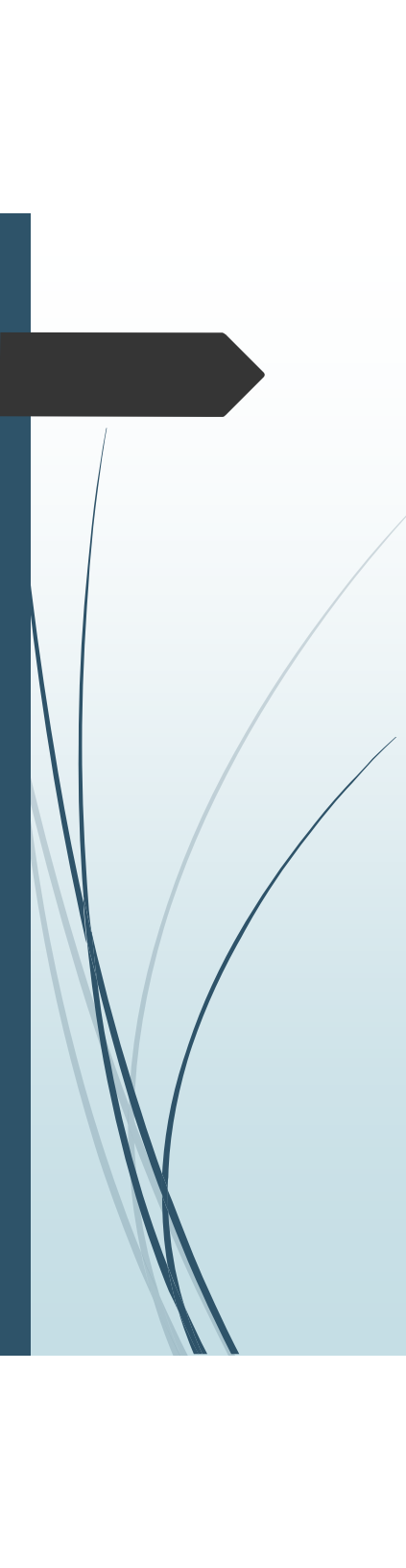
Decision Trees

Decision Trees:

- It is used for both classification and regression tasks.
- Decision Tree Analogy:

```
marks = 25
if (marks < 50):
    print('Fails')
elif (50 < marks < 60):
    print("B Grade")
else:
    print("A Grade")
```





Decision Trees:

► Key Terms:

Root Node: The starting point that represents the entire dataset, which is then divided into two or more subsets.

Internal Node (Decision Node): A node that performs a test on an attribute's value to split the data into further branches.

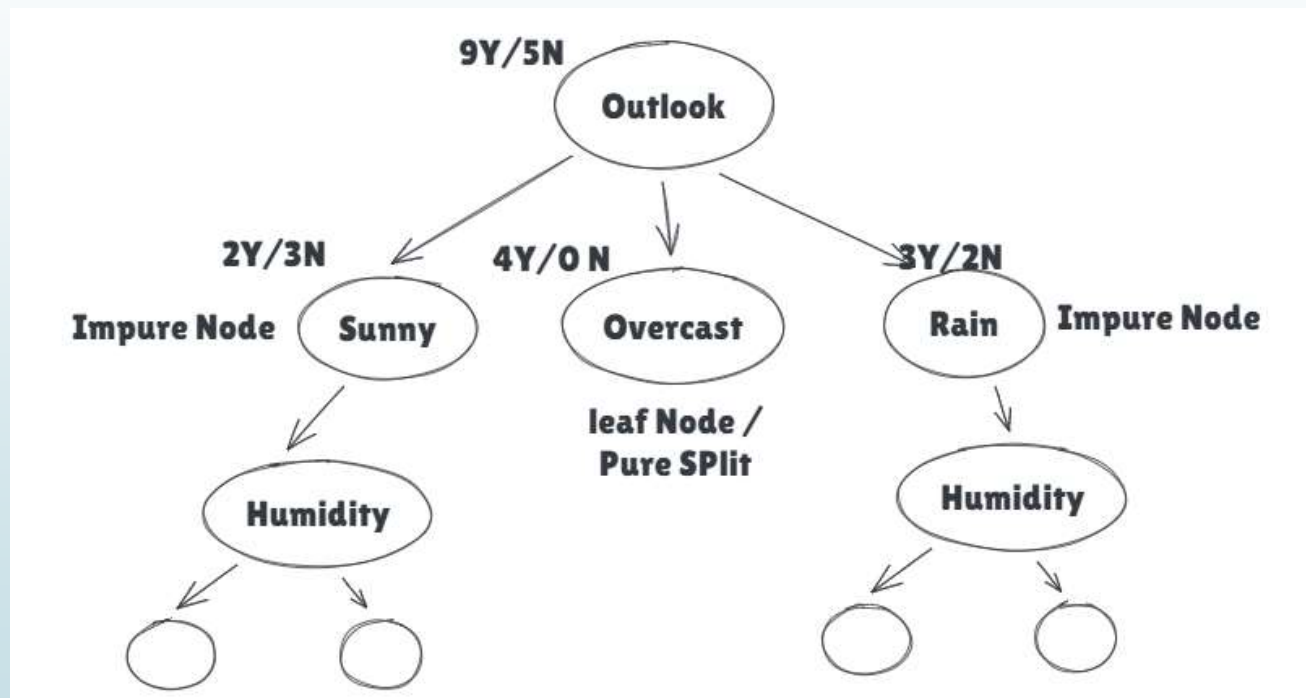
Leaf Node (Terminal Node): The final node that does not split further; it represents the final decision or the class label (e.g., "Yes" or "No", "Spam" or "Not Spam").

Decision Trees:

- Consider a dataset

Outlook	Temperature	Humidity	Windy	PlayTennis
Sunny	Hot	High	False	No
Sunny	Hot	High	True	No
Overcast	Hot	High	False	Yes
Rainy	Mild	High	False	Yes
Rainy	Cool	Normal	False	Yes
Rainy	Cool	Normal	True	No
Overcast	Cool	Normal	True	Yes
Sunny	Mild	High	False	No
Sunny	Cool	Normal	False	Yes
Rainy	Mild	Normal	False	Yes
Sunny	Mild	Normal	True	Yes
Overcast	Mild	High	True	Yes
Overcast	Hot	Normal	False	Yes
Rainy	Mild	High	True	No

Decision Trees:





Decision Trees:

- **Questions:**

- From which feature we have to start splitting?

Information Gain

- How to check that the splitted node is pure or Impure?

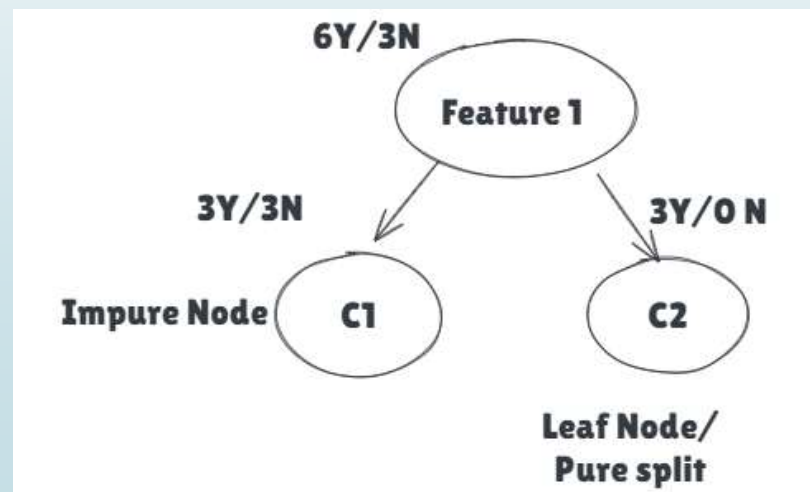
Entropy and Gini Impurity

Decision Trees:

► Entropy:

$$H(S) = - p_{(+)} \log_2 p_{(+)} - p_{(-)} \log_2 p_{(-)}$$

Consider a split;



Decision Trees:

Entropy of C2

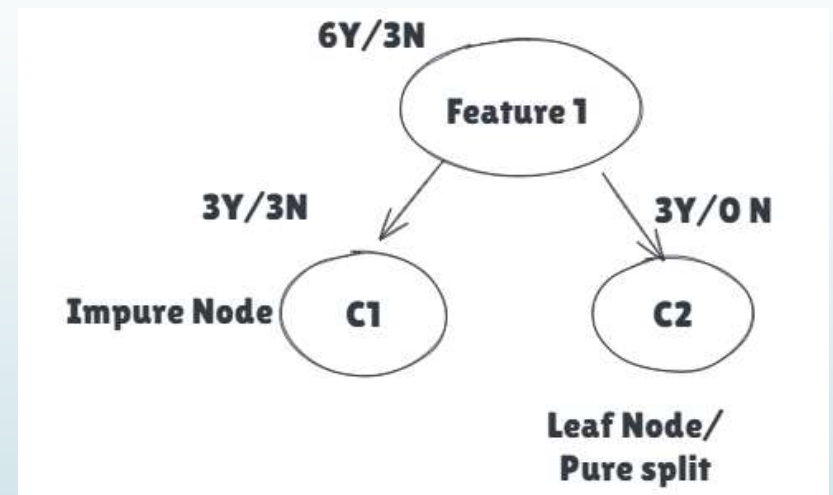
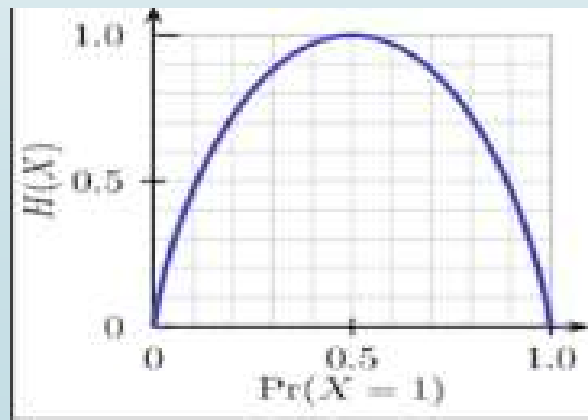
$$H(C_2) = -\frac{3}{3} \log_2 \left(\frac{3}{3} \right) - \frac{0}{3} \log_2 \left(\frac{0}{3} \right) = 0$$

Hence C2 is a pure split

Entropy of C1

$$H(C_1) = -\frac{3}{6} \log_2 \left(\frac{3}{6} \right) - \frac{3}{6} \log_2 \left(\frac{3}{6} \right) = 1$$

Hence C1 is an impure split



Decision Trees:

- Gini Impurity:

$$G = 1 - \sum_{k=1}^K p_k^2$$

K $\Rightarrow$ The number of classes

p_k $\Rightarrow$ Proportion of samples of class k in the node

Decision Trees:

Gini Impurity of C2

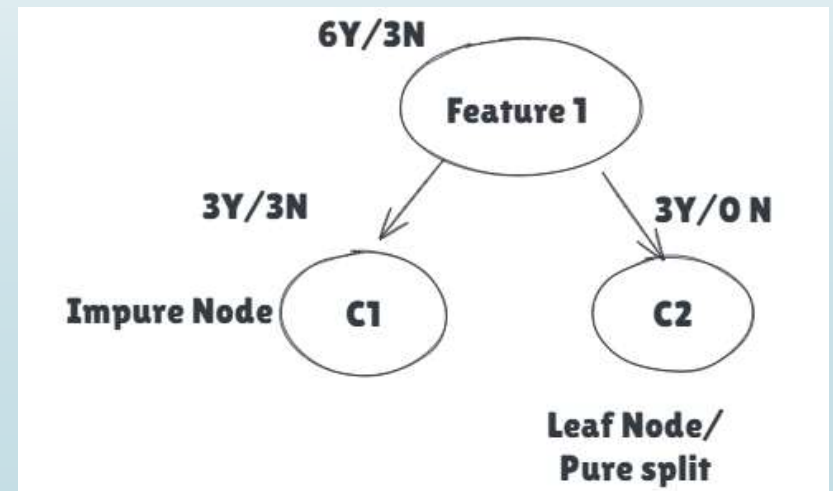
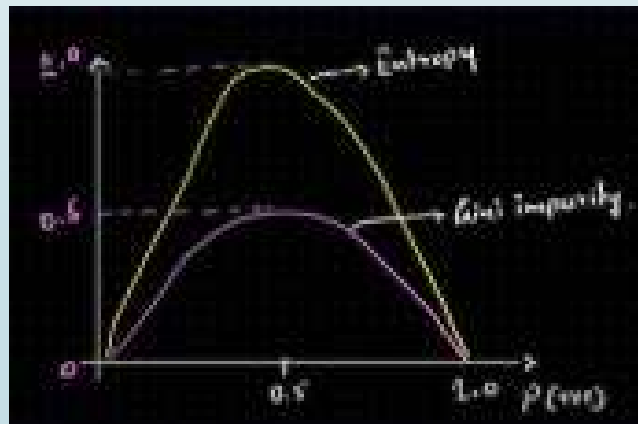
$$GI(C_2) = 1 - (p_+^2 + p_-^2) = 1 - \left(\left(\frac{3}{3}\right)^2 + \left(\frac{0}{3}\right)^2\right) = 1 - 1 = 0$$

Hence C2 is a pure split

Gini Impurity of C1

$$GI(C_1) = 1 - (p_+^2 + p_-^2) = 1 - \left(\left(\frac{3}{6}\right)^2 + \left(\frac{3}{6}\right)^2\right) = 1 - 1/2 = 0.5$$

Hence C1 is an impure split



Decision Trees:

- Information Gain:

$$Gain(S, A) = H(S) - \sum_{v \in \text{Values}(A)} \frac{|S_v|}{|S|} H(S_v)$$

Decision Trees:

Entropy of Root Node

$$H(S) = -p_+ \log_2 p_+ - p_- \log_2 p_- = -\frac{6}{9} \log_2 \left(\frac{6}{9}\right) - \frac{3}{9} \log_2 \left(\frac{3}{9}\right) = 0.27$$

Entropy of C1

$$H(C_1) = -\frac{3}{6} \log_2 \left(\frac{3}{6}\right) - \frac{3}{6} \log_2 \left(\frac{3}{6}\right) = 1$$

Entropy of C2

$$H(C_2) = -\frac{3}{3} \log_2 \left(\frac{3}{3}\right) - \frac{0}{6} \log_2 \left(\frac{0}{6}\right) = 0$$

Then, information gain of feature 1

$$\text{Gain}(S, f_1) = 0.27 - \left[\frac{6}{9} \times 1 + \frac{3}{9} \times 0 \right] = 0.18$$

Similarly, the information gain of feature 2

$$\text{Gain}(S, f_2) = 0.25$$

Decision Trees:

► Decision Tree for numerical Data:

Petal Length	Sepal Length	Type
1.34	0.34	Setosa
3.45	1.45	Versicolor
1.69	0.98	Setosa
2.56	1.79	Virginica
3.00	1.13	Versicolor
1.3	0.88	Setosa

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